Classifications follow the ACMG/AMP Variant Interpretation Guidelines criteria (https://pubmed.ncbi.nlm.nih.gov/25741868/) with the following adjustments by the Sequence Variant Interpretation (SVI) Working Group and considerations from the Low-Penetrance/Risk Allele Working Group. Taken from: https://clinicalgenome.org/tools/clingen-variant-classification-guidance/ (accessed March 2026) and https://clinicalgenome.org/working-groups/low-penetrance-risk-allele-working-group/ (accessed March 2026)

Updated recommendation for the benign stand-alone ACMG/AMP criterion

Ghosh R, Harrison SM, Rehm HL, Plon SE, Biesecker LG; ClinGen Sequence Variant Interpretation Working Group.

The Clinical Genome Resource (ClinGen) Sequence Variant Interpretation Working Group set out to refine the American College of Medical Genetics and Genomics and the Association of Molecular Pathologists (ACMG/AMP) variant pathogenicity recommendations for stand-alone rule BA1 (a variant with minor allele frequency [MAF] > 0.05 is benign), by clarifying how it should be used and specifying a set of variants that should be exempted from this rule. We cross-referenced ClinVar and Exome Aggregation Consortium data to identify variants for which there was a plausible argument for pathogenicity and the variant exists in one or more population data sets at MAF > 0.05. We identified nine such variants that were present in these data sets that may not be benign. The ACMG/AMP criteria were applied to these variants that resulted in four pathogenic and five variants of uncertain significance. We have refined benign rule BA1 by clarifying terms used to describe its use, which databases we recommend using, and assumptions made about this rule. We also recognized an initial list of nine variants for which there was some evidence of pathogenicity even though the MAF was high for these variants. We specify processes whereby individuals can petition ClinGen for amendments to our variant-specific assertions and the criteria experts should use when setting a numerically lower threshold for BA1 for specific genes.

Recommendations for interpreting the loss of function PVS1 ACMG/AMP variant criterion

Abou Tayoun AN, Pesaran T, DiStefano MT, Oza A, Rehm HL, Biesecker LG, Harrison SM; ClinGen Sequence Variant Interpretation Working Group (ClinGen SVI).

The 2015 ACMG/AMP sequence variant interpretation guideline provided a framework for classifying variants based on several benign and pathogenic evidence criteria, including a pathogenic criterion (PVS1) for predicted loss of function variants. However, the guideline did not elaborate on specific considerations for the different types of loss of function variants, nor did it provide decision-making pathways assimilating information about variant type, its location, or any additional evidence for the likelihood of a true null effect. Furthermore, this guideline did not take into account the relative strengths for each evidence type and the final outcome of their combinations with respect to PVS1 strength. Finally, criteria specifying the genes for which PVS1 can be applied are still missing. Here, as part of the ClinGen Sequence Variant Interpretation (SVI) Workgroup's goal of refining ACMG/AMP criteria, we provide recommendations for applying the PVS1 criterion using detailed guidance addressing the above-mentioned gaps. Evaluation of the refined criterion by seven disease-specific groups using heterogeneous types of loss of function variants (n = 56) showed 89% agreement with the new recommendation, while discrepancies in six variants (**11**%) were appropriately due to disease-specific refinements. Our recommendations will facilitate consistent and accurate interpretation of predicted loss of function variants.

Application of the ACMG/AMP Framework to Capture Evidence Relevant to Predicted and Observed Impact on Splicing: Recommendations from the ClinGen SVI Splicing Subgroup

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The American College of Medical Genetics and Genomics (ACMG) and the Association for Molecular Pathology (AMP) framework for classifying variants uses six evidence categories related to the splicing potential of variants: PVS1 (null variant in a gene where loss-of-function is the mechanism of disease), PS3 (functional assays show damaging effect on splicing), PP3 (computational evidence supports a splicing effect), BS3 (functional assays show no damaging effect on splicing), BP4 (computational evidence suggests no splicing impact), and BP7 (silent change with no predicted impact on splicing). However, the lack of guidance on how to apply such codes has contributed to variation in the specifications developed by different Clinical Genome Resource (ClinGen) Variant Curation Expert Panels. The ClinGen Sequence Variant Interpretation (SVI) Splicing Subgroup was established to refine recommendations for applying ACMG/AMP codes relating to splicing data and computational predictions. Our study utilised empirically derived splicing evidence to: 1) determine the evidence weighting of splicing-related data and appropriate criteria code selection for general use, 2) outline a process for integrating splicing-related considerations when developing a gene-specific PVS1 decision tree, and 3) exemplify methodology to calibrate bioinformatic splice prediction tools. We propose repurposing of the PVS1_Strength code to capture splicing assay data that provide experimental evidence for variants resulting in RNA transcript(s) with loss of function. Conversely BP7 may be used to capture RNA results demonstrating no impact on splicing for both intronic and synonymous variants, and for missense variants if protein functional impact has been excluded. Furthermore, we propose that the PS3 and BS3 codes are applied only for well-established assays that measure functional impact that is not directly captured by RNA splicing assays. We recommend the application of PS1 based on similarity of predicted RNA splicing effects for a variant under assessment in comparison to a known Pathogenic variant. The recommendations and approaches for consideration and evaluation of RNA assay evidence described aim to help standardise variant pathogenicity classification processes and result in greater consistency when interpreting splicing-based evidence.

SVI Recommendation for De Novo Criteria (PS2 & PM6) - Version 1.1

The Sequence Variant Interpretation (SVI) Working Group proposes a point -based system to determine the

strength of de novo evidence (ACMG/AMP criteria codes PS2 and PM6) based upon three parameters:

- confirmed parental relationships versus assumed parental relationships status
- phenotypic consistency
- number of de novo observations

To determine the appropriate strength level to apply for de novo occurrence(s), each proband with a de novo variant is awarded a point value based upon phenotypic consistency and confirmed or assumed parental relationships (Table 1). The combined point value of all de novo occurrences is then compared to Table 2 to determine the applicable evidence strength level. For example, if a NIPBL

variant was de novo in one patient with Cornelia de Lange syndrome, with confirmed parental relationships (2 points; Table 1) and de novo in two additional unrelated patients with Cornelia de Lange syndrome with unconfirmed parental relationships (1 + 1 points; Table 1), then VeryStrong evidence level is applied (PS2_VeryStrong) based on combined point value of 4 (Table 2). If the parents have not been tested for parentage or for the variant, no points should be awarded.

Table 1. Points* awarded per *de novo* occurrence

Phenotypic consistency	Points per Proband	
	***de novo* with confirmed parental relationships**	***de novo* with unconfirmed parental relationships**
Phenotype highly specific for gene	2	1
Phenotype consistent with gene but not highly specific	1	0.5
Phenotype consistent with gene but not highly specific and high genetic heterogeneity**	0.5	0.25
Phenotype not consistent with gene	0	0

*Note that these points are *not* equivalent to the points used to classify a variant per the Tavtigian et al 2020 "Fitting a naturally scaled point system to the ACMG/AMP variant classification guidelines"
**Maximum allowable value of 1 may contribute to overall score

Table 2. Recommendation for determining the appropriate ACMG/AMP evidence strength level for *de novo* occurrence(s)

Supporting (PS2_Supporting or PM6_Supporting)	**Moderate** (PS2_Moderate or PM6)	**Strong** (PS2 or PM6_Strong)	**Very Strong** (PS2_VeryStrong or PM6_VeryStrong)
0.5	1	2	4

For all uses of de novo criteria, the phenotype in the patient must be consistent with the gene/disease association as recommended in the ACMG/AMP guidelines. When the patient's phenotype is consistent with the gene/disease association but not highly specific, we recommend decreasing the points awarded. For example:

● A patient with early infantile epileptic encephalopathy and a de novo SIK1 variant with confirmed parental relationships is awarded 1 point (as the patient's phenotype is consistent with the gene but not highly specific and the variant is de novo with confirmed parental relationships). If this patient is the only de novo occurrence for the variant, then a Moderate strength level (PS2_Moderate) is applied.

○ If two additional unrelated patients with early infantile epileptic encephalopathy and a de novo SIK1 variant with confirmed parental relationships are identified, then the combined point value is 3 (as each patient is awarded 1 point). For these combined occurrences, a Strong strength level (PS2) is applied as the points reach the Strong threshold (2 points) but not the VeryStrong threshold (4 points).

● A patient with nonsyndromic intellectual disability and a de novo ASH1L variant is awarded 0.5 points (as the variant is de novo with confirmed parental relationships and patient's phenotype is consistent with the gene but not highly specific and there is significant evidence of genetic

heterogeneity). If this patient is the only de novo occurrence for the variant, then a Supporting strength level (PS2_Supporting) is applied.

○ If a second patient with nonsyndromic intellectual disability and a de novo ASH1L variant with confirmed parental relationships is identified, then the combined point value is 1 (as each patient is awarded 0.5 points). For these combined occurrences, a Moderate strength level (PS2_Moderate) is applied.

● A patient with developmental delay but no other features of Cornelia de Lange syndrome and a de novo NIPBL variant with unconfirmed parental relationships is awarded zero points as this phenotype is not consistent with the gene/disease association. If this patient was the only de novo occurrence for the variant, then no de novo criteria are applied. Additional considerations for applying de novo criteria based on inheritance:

● Conditions with X-linked inheritance: if the variant occurs de novo in an unaffected carrier mother, and family history is consistent - i.e., she has no affected brothers/other male relatives apart from her affected son(s) – de novo criteria may be applied despite the fact that she is unaffected.

● Autosomal recessive conditions: for a de novo occurrence in a gene associated with a condition inherited in an autosomal recessive pattern without an additional pathogenic/likely pathogenic variant identified, the strength of evidence should be decreased by one level.

● Mosaicism: for cases with apparent germline mosaicism (multiple affected siblings with both parents negative for the variant), parental relationships must be confirmed in order for de novo criteria to apply.

Recommendations for application of the functional evidence PS3/BS3 criterion using the ACMG/AMP sequence variant interpretation framework

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Background The American College of Medical Genetics and Genomics (ACMG)/Association for Molecular Pathology (AMP) clinical variant interpretation guidelines established criteria (PS3/BS3) for functional assays that specified a “strong” level of evidence. However, they did not provide detailed guidance on how functional evidence should be evaluated, and differences in the application of the PS3/BS3 codes is a contributor to variant interpretation discordance between laboratories. This recommendation seeks to provide a more structured approach to the assessment of functional assays for variant interpretation and guidance on the use of various levels of strength based on assay validation.

Methods The Clinical Genome Resource (ClinGen) Sequence Variant Interpretation (SVI) Working Group used curated functional evidence from ClinGen Variant Curation Expert Panel-developed rule specifications and expert opinions to refine the PS3/BS3 criteria over multiple in-person and virtual meetings. We estimated odds of pathogenicity for assays using various numbers of variant controls to determine the minimum controls required to reach moderate level evidence. Feedback from the ClinGen Steering Committee and outside experts were incorporated into the recommendations at multiple stages of development.

Results The SVI Working Group developed recommendations for evaluators regarding the assessment of the clinical validity of functional data and a four-step provisional framework to determine the appropriate strength of evidence that can be applied in clinical variant interpretation. These steps are: 1. Define the disease mechanism; 2. Evaluate applicability of general classes of assays used in the field; 3. Evaluate validity of specific instances of assays; 4. Apply evidence to individual variant interpretation. We found that a minimum of eleven total pathogenic and benign variant controls are required to reach moderate-level evidence in the absence of rigorous statistical analysis.

Conclusions The recommendations and approach to functional evidence evaluation described here should help clarify the clinical variant interpretation process for functional assays. Further, we hope that these recommendations will help develop productive partnerships with basic scientists who have developed functional assays that are useful for interrogating the function of a variety of genes.

SVI Recommendation for Absence/Rarity (PM2) - Version 1.0

The ClinGen Sequence Variant Interpretation (SVI) Working Group proposes decreasing the weight of criterion PM2 ("Absent from controls, or at extremely low frequency if recessive, in Exome Sequencing Project, 1000Genomes Project, or Exome Aggregation Consortium") from a Moderate strength level to a Supporting strength level (PM2_Supporting). After substantial analysis and modeling, the SVI WG proposes this weight adjustment due to concerns that absence or rarity is given too much weight in the 2015 framework and that this type of evidence does not meet the relative odds of pathogenicity for a Moderate pathogenic evidence, estimated to be 4.33:1 (PMID:29300386). This concern is supported by findings from the Exome Aggregation Consortium (ExAC) database that 99% of identified high-quality variants have a frequency <1%, that 54% of identified high-quality ExAC variants are only seen once in the entire data set, suggesting that rarity is actually common, and that all individuals harbor variants that are absent from the rest of the population (PMID: 27535533). By decreasing the weight of this criterion, this change will potentially impact variant classifications. Therefore the SVI proposes a novel criteria combination not listed in the combining rules outlined in the 2015 ACMG/AMP guideline. We propose that the combination of one Very Strong criterion and one Supporting criterion reach a classification of Likely pathogenic. This combination rule is supported by the Bayesian framework which shows that one Very Strong Pathogenic criterion and one Supporting Pathogenic criterion results in Post_P of 0.988, which falls within the Likely pathogenic range (0.90-0.99). This combining rule will allow novel LoF variants, that reach PVS1 and PM2 reduced to supporting, to still be classified as Likely pathogenic. We anticipate that other adjustments in the relative weight of evidence will be necessary in the future to accommodate this change in PM2 weight.

SVI Recommendation for in trans Criterion (PM3) - Version 1.0

The Sequence Variant Interpretation (SVI) Working Group proposes a point-based system to determine the strength of in trans observations (ACMG/AMP criterion PM3) based upon variant phasing and classification of the variant occurring on the other allele. Additionally, SVI recommends a revision to the criterion definition to indicate this evidence should only be applied if the individual is affected: SVI revision to PM3: For recessive disorders, detected in trans with a pathogenic or likely pathogenic variant in an affected patient To determine the appropriate strength level to apply for in trans occurrence(s), each proband is awarded a point value based upon phasing of the two variants in question (confirmed in trans versus unknown) and classification of the variant on the other allele (Table 1). The combined point value of all proband occurrences is then summed and compared to

Table 2 to determine the applicable evidence strength level. For example, if assessing PAH variant NM_000277.3:c.1208C>T (p.Ala403Val) and the variant was confirmed in trans with Likely pathogenic variant c.1301C>A (p.Ala434Asp) in one proband (1.0 points; Table 1) and confirmed in trans with Pathogenic variant c.331C>T (p.Arg111Ter) in another proband (1.0 points, Table 1), then PM3 at the Strong strength level (PM3_Strong) is applicable (2.0 points total; Table 2).

Table 1. Points awarded per in *trans* proband

Classification/Zygosity of other variant [1]	Points per Proband	
	Confirmed in *trans*	Phase unknown
Pathogenic or Likely pathogenic variant	1.0	0.5 (P) 0.25 (LP)
Homozygous occurrence *(max point 1.0)*	0.5	N/A
Uncertain significance variant *(max point 0.5)*	0.25	0.0

[1]All variants should be sufficiently rare (meet PM2 specification); P - Pathogenic; LP - Likely pathogenic

Table 2. Recommendation for determining the appropriate evidence strength level for PM3

PM3_Supporting	PM3	PM3_Strong	PM3_VeryStrong
0.5	1.0	2.0	4.0

Considerations:

● Allele Frequency - Application of PM3 is contingent on the allele frequency of the variant being assessed and the variant presumably on the other allele both being sufficiently rare (meets PM2 threshold). This contingency is to avoid incorrect application of PM3 to high frequency variants that are likely to occur in trans with P/LP variants based on frequency.

● Phasing - If the phase cannot be determined, it is recommended that at least two different LP/P variants (depending on classifications) are needed to equal the weight of one LP/P co-occurrence confirmed in trans.

○ In confirmation of phasing, if only one parent is tested and found to carry one allele, variants can be counted as in trans. For example, assessing PAH variant c.601C>T (p.His201Tyr) and variant was identified in PKU proband who also carries known pathogenic variant c.734T>C (p.Val245Ala). Only the mother is available for testing and the mother only carries c.734T>C (p.Val245Ala) variant, then variants can be considered in trans.

● Classification - Probands should be weighted less when the variant on the other allele is of uncertain significance and rare (meets PM2); however, weight may vary by gene size as larger genes are more likely to have a second variant by chance (default 0.25 points). If the variant on the other allele is classified as P or LP, weighting depends on phasing (see Phasing above), with P/LP being weighted equally if confirmed in trans and different point values per proband if phasing is unknown (0.5 points and 0.25 points, respectively). To avoid circularity, in all instances (phasing confirmed or unknown), the classification of the variant on the other allele should not use evidence from the variant being interrogated.

● Homozygous occurrences - For homozygous occurrences, the default weight is dropped to 0.5 points, as a rare homozygous occurrence may be due to consanguinity. A recommended max of 1.0 points of all homozygous cases is suggested to prevent overclassification of homozygous occurrences in the absence of additional data.

ClinGen guidance for use of the PP1/BS4 co-segregation and PP4 phenotype specificity criteria for sequence variant pathogenicity classification

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The 2015 American College of Medical Genetics and Genomics and the Association for Molecular Pathology variant classification publication established a standard employed internationally to guide laboratories in variant assessment. Those recommendations included both pathogenic (PP1) and benign (BS4) criteria for evaluating the inheritance patterns of variants, but details of how to apply those criteria at appropriate evidence levels were sparse. Several publications have since attempted to provide additional guidance, but anecdotally, this issue is still challenging. Additionally, it is not clear that those prior efforts fully distinguished disease-gene identification considerations from variant pathogenicity considerations nor did they address autosomal-recessive and X-linked inheritance. Here, we have taken a mixed inductive and deductive approach to this problem using real diseases as examples. We have developed a practical heuristic for genetic co-segregation evidence and have also determined that the specific phenotype criterion (PP4) is inseparably coupled to the co-segregation criterion. We have also determined that negative evidence at one locus constitutes positive evidence for other loci for disorders with locus heterogeneity. Finally, we provide a points-based system for evaluating phenotype and co-segregation as evidence types to support or refute a locus and show how that can be integrated into the Bayesian framework now used for variant classification and consistent with the 2015 guidelines.

Calibration of computational tools for missense variant pathogenicity classification and ClinGen recommendations for PP3/BP4 criteria

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Recommendations from the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) for interpreting sequence variants specify the use of computational predictors as "supporting" level of evidence for pathogenicity or benignity using criteria PP3 and BP4, respectively. However, score intervals defined by tool developers, and ACMG/AMP recommendations that require the consensus of multiple predictors, lack quantitative support. Previously, we described a probabilistic framework that quantified the strengths of evidence (supporting, moderate, strong, very strong) within ACMG/AMP recommendations. We have extended this framework to computational predictors and introduce a new standard that converts a tool's scores to PP3 and BP4 evidence strengths. Our approach is based on estimating the local positive predictive value and can calibrate any computational tool or other continuous-scale evidence on any variant type. We estimate thresholds (score intervals) corresponding to each strength of evidence for pathogenicity and benignity for thirteen missense variant interpretation tools, using carefully assembled independent data sets. Most tools achieved supporting evidence level for both pathogenic and benign classification using newly established thresholds. Multiple tools reached score thresholds justifying moderate and several reached strong evidence levels. One tool reached very strong

evidence level for benign classification on some variants. Based on these findings, we provide recommendations for evidence-based revisions of the PP3 and BP4 ACMG/AMP criteria using individual tools and future assessment of computational methods for clinical interpretation.

The ACMG/AMP reputable source criteria for the interpretation of sequence variants

Leslie G Biesecker, Steven M Harrison; ClinGen Sequence Variant Interpretation Working Group

To the Editor: In 2015, the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) promulgated recommendations for the interpretation of sequence variants.[1] These recommendations have been widely implemented and shown to be useful for improving variant classification consistency.[2–4] From the beginning, they were recognized to be a starting point for further future refinements and extensions. The Clinical Genome (ClinGen) Resource is focused on curating the genome for use in molecular diagnosis.[5] One such effort is the Sequence Variant Interpretation Working Group, which has taken on the task of refining and evolving the current ACMG/AMP recommendations. This working group meets regularly and also, as individuals, interacts widely with the clinical testing community. Through these interactions, the working group has received input from multiple sources that two related criteria in the original recommendations should be considered for removal from the ACMG/AMP framework:

- PP5 “Reputable source recently reports variant as pathogenic, but the evidence is not available to the laboratory to perform an independent evaluation.”
- BP6 “Reputable source recently reports variant as benign, but the evidence is not available to the laboratory to perform an independent evaluation.”

It is our strongly held view that primary data are far preferable to expert opinion without access to primary data. The PP5 and BP6 criteria rely on assertions that are not directly linked to the evidence on which they were based. These criteria might have been appropriate in 2015 as they were originally intended as a bridge to allow the community to benefit from clinical laboratory expertise and locus-specific research databases, prior to the wider use of resources such as ClinVar that provide mechanisms for laboratories and researchers to share underlying primary data. Indeed, ClinVar has been even more successful than hoped and now includes 580,831 assertions for over 375,106 variants (ClinVar website accessed 8 January 2018). Submissions with “assertion criteria provided” review status, designating that the submitter either provided their evidence to ClinVar or indicated a willingness to provide evidence upon inquiry, account for 81% (470,245/580,831) of all submissions to ClinVar (ClinVar website accessed 8 January 2018). Therefore, there is less need to rely on assertions from reputable sources in the absence of primary evidence.

A second rationale for these two criteria was to support the efforts of the Sharing Clinical Reports Project (https://www.clinicalgenome.org/data-sharing/sharing-clinical-reports-project-scrp/), in which clinicians collected the test reports (including variant interpretation) produced by a large commercial laboratory that for the past decade has consistently declined to share underlying data or to submit assertions to ClinVar. As data for hereditary breast and ovarian cancer susceptibility alleles have increasingly been forthcoming from other laboratories, the necessity of this secondary information has declined and the currency of these data has receded.

Finally, we are concerned that these two criteria may be commonly misused by laboratories that incorporate primary data into variant assessment (e.g., functional data, criteria PS3 and BS3) and at the same time invoke criteria PP5 and BP6 for existing classifications that are based on the same set of data, which may lead to double counting, and potentially lead to errors in classification.

Based on these considerations, we propose that laboratories discontinue the use of criteria PP5 and BP6 as soon as that is practically achievable. We have removed these criteria from the ClinGen Variant Curation Interface. However, as with all types of evidence, interpretation of variants is the responsibility of the clinical testing laboratory director, who should account for the entirety of evidence and the sources of the data, and these recommendations should not be interpreted otherwise.

Low Penetrance/Risk Allele

Low penetrance / risk alleles are assessed based on the the Low-Penetrance/Risk Allele Working Group which aims to develop a consensus on the terminology needed to categorize both risk alleles and low-penetrance Mendelian variants and to develop a standardized classification framework to evaluate these types of variants. We follow the recommendations stated in https://clinicalgenome.org/docs/current-challenges-in-the-clinical-classification-of-low-penetrance-variants/ (accessed 31.03.2026) and https://doi.org/10.1016/j.gim.2023.101036:

Recommendations for Risk Allele Evidence Curation, Classification, and Reporting from the ClinGen Low Penetrance/Risk Allele Working Group

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Abstract

Purpose: Genetic variants at the low end of the penetrance spectrum have historically been challenging to interpret since their high population frequencies exceed the disease prevalence of the associated condition, leading to a lack of clear segregation between the variant and disease. There is currently substantial variation in the classification of these variants, and no formal classification framework has been widely adopted. The Clinical Genome Resource Low Penetrance/Risk Allele Working Group was formed to address these challenges and promote harmonization within the clinical community.

Methods: The work presented here is the product of internal and community Likert-scaled surveys in combination with expert consensus within the Working Group.

Results: We formally recognize risk alleles and low penetrance variants as distinct variant classes from those causing highly penetrant disease that require special considerations regarding their clinical classification and reporting. First, we provide a preferred terminology for these variants. Second, we focus on risk alleles and detail considerations for reviewing relevant studies and present a framework for the classification these variants. Finally, we discuss considerations for clinical reporting of risk alleles.

Conclusion: These recommendations support harmonized interpretation, classification, and reporting of variants at the low end of the penetrance spectrum.

Current challenges in the clinical classification of low penetrance variants

Ryan J. Schmidt, Marcie Steeves, Pinar Bayrak-Toydemir, Alicia Byrne, Bryony Thompson, Mythily Ganapathi, Melissa Kelly, Deqiong Ma, Erik Puffenberger, Amanda Spurdle, and Matthew Lebo on behalf of the ClinGen

Low Penetrance/Risk Allele Working Group

Definition of Low Penetrance Variants

Prior work by the ClinGen Low Penetrance/Risk Allele Working Group established terminology to describe variants at the low end of the penetrance spectrum (Figure 1)1. Penetrance, here, refers to absolute penetrance, i.e., the probability of disease manifestations occurring given that an individual carries a certain variant. Absolute penetrance is not a measure that is relative to the penetrance of other variants in a certain gene. The recommendations of the Working Group introduced two variant classes - risk alleles and low penetrance variants - distinguished from typical Mendelian high penetrance variants. This work defined risk alleles as having: "appreciable allele frequencies in the population given that they do not cause disease with a high penetrance. These variants are unlikely to segregate with disease in families with a clear inheritancepattern as they explain only a portion of disease risk. While additional environmental and genetic factors affect the penetrance of disease for both low penetrant variants and risk alleles, it is likely that their contribution to the overall disease risk is greater with risk alleles. However, risk alleles still convey enough risk such that their presence or absence may be considered clinically significant.1" The work further defined a low penetrance variant as variants that: "may also show some evidence of segregation with disease in families. However, the penetrance associated with these variants is reduced such that the presence of the variant alone does not determine whether a disease will occur.1" Additionally, a community survey performed by our working group suggested that: "For low penetrance variants, there was agreement with the lower bound of penetrance being in the 10–20% absolute risk range and the upper bound being in the 50% range1"

These variant classes are currently defined qualitatively and the absolute penetrance thresholds between these classes have not yet been firmly established. However, preliminary community survey results suggest that most consider low penetrance variants to be associated with an absolute penetrance range between ~10-20% on the low end and up to ~50% on the high end. In contrast, risk alleles are considered to have an absolute penetrance below ~10-20%. Further, ClinGen's Variant Curation Expert Panels (VCEPs) are recommended to use the online allele frequency calculator2 for setting thresholds for allele frequency evidence assignment. In this tool, 20% penetrance has been set implicitly within the calculation of maximum allowable allele frequency, which aligns with the lower bound for absolute penetrance for low penetrance variants that our working group has preliminarily identified. Data quantifying penetrance is not currently available for most individual variants, especially those with low population frequencies; however, penetrance estimates are available in aggregate for specific variant types for some genes. Thus, it may be difficult to determine whether a variant of interest should be considered as a low penetrance variant. Novel variants or variants lacking sufficient genetic evidence are not candidates to be classified as low penetrance variants at this time, unless all pathogenic variants in a given gene are considered low penetrance (see "Low Penetrance Gene" section below). We anticipate that evidence of variant penetrance may become more available in the future as a result of large-scale population studies that include detailed clinical phenotyping. Nevertheless, there are variants whose available evidence suggests that they

are clearly low penetrance where they have some genetic evidence supporting pathogenicity but can also be found at an appreciable rate in unaffected individuals. For these variants, the standard variant classification frameworks that assume high penetrance consistent with monogenic disease presentation3,4 can lead to misclassification due to conflicting evidence codes being applied or limited evidence strength. Refinement of the standard frameworks is therefore necessary to enable proper classification of these low penetrance variants. Below we highlight the challenges surrounding the classification of low penetrance variants. Future work will be required to develop or modify a variant classification framework to better accommodate low penetrance variants. Current Challenges with Low Penetrance Variants Currently, low penetrance variants frequently share similar types of evidence of pathogenicity with high penetrance variants, as opposed to traditional case-control studies used to evaluate risk alleles. Risk alleles are classified primarily based on case-control studies1, whereas low penetrance variants share similar types of evidence of pathogenicity with high penetrance variants including case counts, population frequency data, computational prediction data, functional data, and segregation data4. However, discussions within our working group suggest that low penetrance variants may receive inaccurate classifications using existing frameworks for high penetrance variants due to 1) evidence that does not fit into existing definitions, 2) conflicting evidence, 3) inappropriate evidence strength calibration. Here we seek to outline the current challenges associated with the classification of low penetrance variants. Defining the Phenotype Low penetrance variants are frequently thought of as variants in a high penetrance disease gene that lead to a similar range of phenotypic features but with a lower risk of disease in individuals with the variant. However, several other scenarios are possible, including: 1) low penetrance variants may be associated with a subset of features observed for high penetrance disease-causing variants in the same gene, 2) low penetrance variants may cause disease at later ages (different age-related penetrance), and 3) low penetrance variants may be associated with a range of features that is distinct from that observed for high penetrance variants in the same gene.Even though penetrance and expressivity are distinct concepts, low penetrance variants may also exhibit variable expressivity among individuals. Variable expressivity refers to different manifestations among individuals with a genetic disease, either in severity or range of phenotypic features. Differences in expressivity, which can also be seen for high penetrance genes, may occur between individuals sharing the same or different alleles in a given gene. The ultimate phenotypic manifestations of low penetrance variants can range from isolated laboratory abnormalities or subclinical phenotypes to full clinical presentation of a recognized disorder. Low penetrance variants can, in some cases, exclusively cause a distinct disorder from high penetrance variants in the same gene that is recognized as a separate named disease entity. ClinGen has issued guidance directing the definition of disease entities, which ultimately guides variant curation5. Following this guidance, we note that separate variant specifications would apply in the case of two distinct gene-disease relationships. Finally, when evaluating association study evidence for low penetrance variants, the definition of cases and controls and methods of phenotyping should be carefully evaluated, especially in light of the considerationslisted above1.

Low Penetrance Genes

Many monogenic diseases are caused by high penetrance pathogenic alleles, with low penetrance variantsen countered infrequently. In contrast, other monogenic diseases are themselves characterized by low penetrance, such that all pathogenic alleles in the gene are associated with low penetrance. These genes may be referred to as “low penetrance genes” (e.g., CHEK2)6–8. Variants in low penetrance genes may be easier to classify as “pathogenic, low penetrance” in the current framework since the maximum allowable populationfrequency threshold is based on variants with

low penetrance. However, the classification of low penetrance variants, even in low penetrance genes, may be impacted by other elements in a framework designed for high penetrance variants.

Population Frequency

In the classification of high penetrance variants, a population frequency that is “too high” relative to the prevalence of the disorder is considered as a strong or very strong benign evidence criterion. Specifically, the BA1 criteria has been defined as “allele frequency is >5%” with some exceptions noted4,9, and BS1 as “allele frequency is greater than expected for disorder”4. However, low penetrance variants are more likely to be found in population studies as not all individuals harboring the variant are affected. Further, guidance from ClinGen VCEPs for assessing specific diseases frequently sets the maximum allowable population frequency thresholds for high penetrance variants encountered4,10. The practice of applying benign criteria to variants for having an appreciable frequency in population studies constrains the classification of low penetrance variants, and alternative classification frameworks may be more appropriate. In the forthcoming update for the sequence variant classification standards11, population frequency will only contribute negative points towards pathogenicity, and variants with appreciable population frequencies will be ineligible to receive points favoring pathogenicity from case counting. The phenomenon of elevated population frequency for some low penetrance variants frequently leads to conflicting evidence when there is genetic evidence of pathogenicity for a low penetrance variant that also presents with sufficient population frequency to meet benign evidence criteria. To resolve the conflicting interpretations frequently associated with low penetrance variants, the working group proposes that benign criteria resulting from higher-than-expected population frequencies should be discounted in cases of low penetrance variants with sufficiently strong genetic evidence of pathogenicity. The strength of evidence required for this change has not yet been determined and would be the subject of future work. For variants where well-designed association studies have been performed showing that the variant is associated with disease, the frequency of the variant in an unaffected population has already been appropriately considered in the context of the study.

Proband Counting

Proband counting refers to the counting of affected individuals that carry a variant of interest, considering the specificity of the phenotype of interest. This may be performed using reports from the published literature or from unpublished cohorts. Proband counting can serve as evidence of pathogenicity for very rare variants when a variant’s population allele frequency is zero or close to zero4. In general, low penetrance variants may be identified in unaffected individuals as the population allele frequency is often appreciable. Of note, the BS2 criterion (observation of a variant in a healthy individual) is not applicable when the disease under consideration is not fully penetrant and should not be applied for low penetrance variants4. Several ClinGen VCEPs have adopted specifications that prohibit proband counting data from being used toprovide pathogenicity evidence when the variant is present in population studies and the PM2 evidence criterion cannot be applied4,12,13. Thus, low penetrance variants are largely ineligible to receive evidence of pathogenicity through proband counting. We anticipate that case-control studies will play a larger role to provide evidence of pathogenicity for low penetrance variants. Cohort-specific calibration may be needed in order to avoid erroneous conclusions due to ascertainment bias for phenotypes that are common in the general population14, and for distinctive phenotypic presentation (cancer types or family history) that are used as clinical characteristics to promote testing of limited gene panels15.

Semi-dominance

For some low penetrance variants, the presence of these variants in the biallelic state leads to high penetrance disease that may be more severe or have earlier onset. This effect has been described as

semi-dominance16. Counting of affected individuals with a variant present in the biallelic state for certain genes in a manner similar to invoking the PM3 criterion4 may serve as evidence of pathogenicity for the variant to serve as a low penetrance variant in the monoallelic state. This method is applicable when the molecular mechanism of disease is similar for both the monoallelic and biallelic disorders. For example, monoallelic variants in the APOB gene are associated with familial hypercholesterolemia with reduced penetrance relative to the biallelic form of the disorder17,18.

Cosegregation

Cosegregation analysis is based on determining the probability that the pattern of cosegregation of a variant in individuals with disease within family pedigrees could be seen by chance. For low penetrance variants, this approach is challenged by several issues. First, for low penetrant variants, fewer variant carriers are expected to be affected, so simple meiosis countingmethods need to account for the fact that larger pedigrees are needed to achieve the same strength of evidence that could be achieved for a high penetrance variant according to the methodology of recent ClinGen Sequence Variant Interpretation working group guidance19. Similar to cosegregation analyses involving high penetrance variants, the presence of an affected family member that does not carry the variant would represent a non-segregation event while an affected individual with the variant represents a segregation event. As detailed above, it is important to consider the definition of phenotype (see above) for low penetrance variants and note that unaffected individuals may have subclinical phenotypes. While segregation data from large pedigrees can provide additional evidence for rare diseases, as with high penetrance variants, it is critical to consider the possibility of phenocopies for common conditions such as cancer or cardiovascular disease. Second, even for more sophisticated approaches that consider gene-specific disease penetrance against population incidence rates, age-related phenocopies, and other features such as sex-specific presentation and variable expression, low penetrance variants have particular challenges. Advanced cosegregation analyses have previously been used in the context of a variant associated with ~50% of the level of risk of the “standard” pathogenic BRCA1 variant20 and adopted for use by VCEPs for multiple cancer predisposition genes. However, these approaches21 require a large amount of data (number and/or extent of pedigrees) and are not robust enough to detect increased risk at the lower ends of the penetrance scale. Specifically, for variants with lower than ~2-fold disease risk, the approach fails and should not be used since it has the potential to provide false evidence for and against pathogenicity21, which is critical for low penetrance genes (e.g., ATM or CHEK2).

Computational Prediction

Missense and Splice-site Variant Effect Predictors Computational prediction has traditionally served as “supporting” evidence in the classification of high penetrance variants, specifically for the classification of missense variants but also splice-impacting variants and in-frame insertion-deletion variants4. Additionally, it has been recommended that in order to apply the evidence criterion for missense variants affecting the same residue as a known pathogenic variant, the candidate missense variant must have an equal or greater computational prediction score than the pathogenic variant22. More recently, meta-predictors and other tools with increased prediction accuracy have been developed; calibration of certain computational prediction tools has been performed indicating that certain score thresholds from these tools provide increased predictive power for suggesting that a variant is pathogenic23. Therefore, assignment of increased evidence strength (beyond the supporting level) for computational prediction tools has been recommended. It is currently unclear if low penetrance variants as a group are associated with lower scores from computational prediction tools relative to high penetrance variants in the same gene(s). Low penetrance variants may have been excluded from training sets during tool development as they might not have been regarded as

definitively pathogenic or benign. If low penetrance variants tend to have lower computational prediction scores, they might require other evidence in order to classify them as pathogenic in current frameworks. Some computational prediction tools incorporate population frequency data24. Given that low penetrance variants may have higher population frequencies, they would receive lower scores from these tools. Finally, certain low penetrance variants may cause disease with low penetrance because they have only partially disrupted splicing or protein function which could feasibly be associated with intermediate score outputs from computational prediction tools, though these scores are not typically designed to detect these nuances.

Protein Domains Enriched for Pathogenic Variants

Enrichment of pathogenic variants within a functional domain serves as an evidence criterion favoring pathogenicity for novel variants identified within the same domain4. Additionally, alteration of critical residues within well-characterized repeated motifs provides evidence of pathogenicity for novel variants predicted to affect these residues25. There are examples of genes where pathogenic variants affecting a specific domain are known to be associated with low penetrance relative to the variants affecting other regions of the protein.For the NOTCH3 gene, cysteine-altering variants at positions involved in disulfide bonding within distal EGF- like repeats are associated with lower penetrance and later age-of-onset for cerebral arteriopathy, autosomal dominant, with subcortical infarcts and leukoencephalopathy (CADASIL)26,27.Thus, novel cysteine-altering variants in these regions are anticipated to cause disease with low penetrance. Given that proteins may contain multiple functional domains that each have a separate impact on disease when altered, it should be considered that alterations in different functional domains might lead to different levels of penetrance. While these “low penetrance domains” may not be evident from review of currently available information, large-scale studies associating genotype and disease risk have the potential to allow them to become more easily identified.

Functional Studies (Experimental Evidence)

Functional studies can be a source of strong evidence of pathogenicity for high penetrance variants4. Given that low penetrance variants may act through the same biological mechanism as high penetrance variants in the same gene, they may show evidence of impact on function in studies traditionally used for high penetrance variants in the same gene. However, the functional impact of low penetrance variants may be reduced in magnitude compared to high penetrance variants. Variants with partial alteration in function are frequently referred to as hypomorphic. Partial loss of function may occur at the RNA-level through altered splicing or at the protein-level through partial functional activity of an expressed protein product. However, not all hypomorphic variants have low penetrance and not all low penetrance variants are hypomorphic. Low penetrance variants may impact only some aspects of protein function, and therefore, may only be detected in some assay types (PALB228 and BRCA1 R1699Q20,29. As a result, proper assay selection and calibration are critical when evaluating low penetrance variants. Variants with intermediate impact, which may be low penetrance variants, have mostly been excluded when calibrating functional assays to date. Further, it is possible that low penetrance variants were inappropriatelyincluded for both benign or pathogenic reference sets used for assay calibration. Thus, low penetrance variants likely were not evaluated independently from benign and high penetrance pathogenic variants for these functional studies. Future well-calibrated functional studies for many genes are needed to completely assess the functional impacts of low penetrance variants, where in an ideal scenario, a well-calibrated assay could reliably predict when a variant is low penetrance based on the result of the functional assay alone, although this may not be possible for all genes.

Low Penetrance Variants in Recessive Scenarios

Our working group considered several scenarios involving either monoallelic or biallelic variants in recessive disease genes that may result in low penetrance. In these examples, the disease phenotype is determined by a combination of variants in trans.

1. Low Penetrance Disease due to Low Penetrance Variant in trans with Fully Penetrant Variant (Homozygosity for the Low Penetrance Variant typically does not lead to disease manifestations)

In certain high penetrance recessive disease genes, compound heterozygosity between a “stronger” null allele and a “weaker” variant that is associated with reduced function causes the disease phenotype with lower penetrance. For example, deficient biotinidase enzyme activity results from biallelic loss of BTD gene function. The BTD NM_001370658.1:c.1270G>C p.Asp424His variant represents a hypomorphic allele with approximately 50% residual enzyme activity that causes disease with low penetrance when in trans with a null allele; however, in homozygosity this variant is not expected to lead to a clinical phenotype30.

2. Low Penetrance Disease due to Low Penetrance Variant Homozygote

In some cases, low penetrance disease is caused by a low penetrance variant carried in the homozygous state. The GJB2 gene has two frequently encountered missense variants [c.101T>C (p.Met34Thr) and c.109G>A (p.Val37Ile)] that result in low penetrance mild hearing loss in homozygotes31. Although individuals who are compound heterozygous for one of these alleles and a typical pathogenic variant also have mild hearing loss, the penetrance for these compound heterozygotes is much higher than the homozygotes.

3. Low Penetrance Disease due to Low Penetrance Gene with Recessive Inheritance

It remains a possibility that for a gene with recessive inheritance, all biallelic combinations of pathogenic variants cause disease with low penetrance similar to what is described in the Low Penetrance Genes section (above).

Related Terminology and Concepts

In the course of our work on low penetrance variants, the ClinGen Low Penetrance/Risk Allele Working Group discussed several terms and concepts with varying degrees of relationship to the concept of low penetrance that are frequently used in the genetics space. As the usage of these terms can vary in the community, we define them here in an unofficial capacity in order to convey how these terms were considered in our discussions.

● Hypomorph

○ Variant with reduced function as measured by a functional assay or by RNA expression analyses. See Functional Studies section for more details.

● Pseudodeficiency

○ Variants that are not pathogenic but appear to have disrupted function based on laboratory

studies that may be performed under non-physiological conditions that do not fully recapitulate human biology. Thus, the ability to precisely measure the lower threshold of activity that leads to cellular and tissue metabolic dyshomeostasis may not be possible.

○ Note that a working group within ClinGen has been formed to address issues surrounding pseudodeficiency alleles (ClinGen Pseudodeficiency Working Group)32.

● Variable Expressivity

○ Variability in the manifestation of disease in terms of severity, age of onset and presence of phenotypic features.

○ Note that individuals with variants exhibiting variable expressivity may not always be identified as having disease depending upon the extent of phenotyping performed, and thus, appear non-penetrant.

● Recessive Variants in Dominant Disease

○ Some high penetrance disorders with dominant inheritance can be caused by a weaker variant in the same gene in homozygosity, which does not cause disease by itself when present in heterozygosity.

○ These recessive variants are not Low Penetrance Variants as they only cause disease with high

penetrance when present in the biallelic state.

● Modifier Variant

○ A variant that does not lead to disease on its own but can increase or decrease the severity of a disease condition caused by a primary disease-causing variant.

○ Examples include MYH7 loss-of-function variants in trans with pathogenic MYH7 gain-of- function alleles33 and certain CFTR tandem repeat length alleles34.

● X-linked Inheritance in Females

○ For female carriers of X-linked recessive disorders such as RPGR-associated retinal dystrophy, a subset of individuals may manifest disease influenced in part by X-chromosome inactivation variability35.

● Manifesting Carriers

○ This scenario results from heterozygous carrier states involving high penetrance pathogenic variants for recessive disease that exhibit mild phenotypic manifestations in the carrier state that are often very low penetrance and lead to clinical manifestations in a subset of individuals in the carrier state.

○ These variants are not Low Penetrance Variants as they cause disease with high penetrance when present in the biallelic state.

○ One example of manifesting carriers can be seen in some individuals with limited phenotypic manifestations who have sickle cell trait, which refers to the carrier state for autosomal recessive sickle cell anemia36